

1 MLE of σ^2 from Multi-variate Normal

Likelihood of the Multi-variate normal:

$$L(\underset{\sim}{\beta}, \sigma^2 | \underset{\sim}{y}) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})}$$

We can then write the log-likelihood:

$$l(\underset{\sim}{\beta}, \sigma^2 | \underset{\sim}{y}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{1}{2\sigma^2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})$$

We then take the partial derivative with respect to σ^2 and set equal to 0:

$$\begin{aligned} \frac{\partial l}{\partial \sigma^2} &= 0 - \frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta}) = 0 \\ \implies n\hat{\sigma}^2 &= (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta}) \\ \implies \hat{\sigma}^2 &= \frac{(\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})}{n} \end{aligned}$$

The $(\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})$ is actually the same as the SSE, so this can be rewritten as:

$$\hat{\sigma}^2 = \frac{SSE}{n}$$

2 MLE of $\underset{\sim}{\hat{\beta}}$ from Multi-variate Normal

$$L = (2\pi)^{-n/2} (\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})}$$

$$l = -\frac{n}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma|) - \frac{1}{2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top \Sigma^{-1} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})$$

$$\begin{aligned} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top \Sigma^{-1} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta}) &= \underset{\sim}{y}^\top \Sigma^{-1} \underset{\sim}{y} - \underset{\sim}{y}^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\beta} - (\underset{\sim}{x}\underset{\sim}{\beta})^\top \Sigma^{-1} \underset{\sim}{y} + (\underset{\sim}{x}\underset{\sim}{\beta})^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\beta} \\ &= \underset{\sim}{y}^\top \Sigma^{-1} \underset{\sim}{y} - 2\underset{\sim}{\beta}^\top \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} + \underset{\sim}{\beta}^\top \underset{\sim}{x}^\top \Sigma^{-1} (\underset{\sim}{x}\underset{\sim}{\beta}) \end{aligned}$$

$$\frac{\partial}{\partial \underset{\sim}{\beta}} - \frac{1}{2} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta})^\top \Sigma^{-1} (\underset{\sim}{y} - \underset{\sim}{x}\underset{\sim}{\beta}) = -\frac{1}{2} [0 - 2\underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} + 2\underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\beta}] = \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} - \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\beta}$$

Set it equal to 0:

$$\begin{aligned} 0 &= \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} - \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\hat{\beta}} \\ \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{x}\underset{\sim}{\hat{\beta}} &= \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} \\ \underset{\sim}{\hat{\beta}} &= (\underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{x})^{-1} \underset{\sim}{x}^\top \Sigma^{-1} \underset{\sim}{y} \end{aligned}$$